

"Along with domestic RTS installations, there is also an aligned goal to solarise government buildings. If rooftop solar is installed on every such building, more than 32 GB can be installed"

–SAPTAK GHOSH, *Senior policy specialist, Centre for Study of Science, Technology and Policy*



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resulted in revenue of Rs 200 crore for these households and savings of nearly Rs 2,000 crore in reduced electricity bills, according to Desai.

The deployment challenge

So far, the biggest resistance came from discoms which, already burdened with losses, were concerned about losing customers. But as solar power has become cheaper, discoms can now procure electricity from RTS at a cheaper rate than they do from other sources. The PM-Surya Ghar: Muft Bijli Yojana has set aside Rs 4,950 crore towards incentives for discoms. "Installation of solar rooftop plants makes consumers self-sustainable at least during daytime by generating electricity and selling the excess power generated, reducing electricity cost by 33 per cent," says Mahantesh Bilagi, managing director of Bengaluru Electricity Supply Company (BESCOM), the nodal agency for residential RTS in Karnataka. "The installation of rooftop plants under the scheme will help discoms in future to manage the gap between demand and supply, as there is increase in power demand year on year." As of mid-September, there were 436,016 registrations in Karnataka under PM-Surya Ghar. Karnataka has a free electricity scheme called Gruha Jyothi for households consuming up to 200 units per month, but there are 3-3.5 million consumers who haven't availed it, Bilagi points out. In Bengaluru, coverage of Gruha Jyothi is about 65 per cent; the remaining households are either ineligible or have opted out. These are the consumers that BESCOM is targeting for RTS. Simultaneously, a survey of government buildings is underway in Bengaluru to assess the potential for installing RTS on them.

Overall, the Muft Bijli Yojana has three main components, explains Ghosh. Along with the subsidy for domestic RTS installations for households, there is also an aligned goal to solarise government buildings. "If rooftop solar is installed on every government building (state and central), more than 32 GW can be installed," says Ghosh. The third component is a sub-scheme for creating model solar villages on the lines of Modhera in Gujarat which in 2022 became India's first village to go 100 per cent solar. A competitive scheme which promises to award Rs 1 crore to one village per district in India based on their success in becoming self-reliant for energy needs, the model solar village programme has an outlay of Rs 800 crore. CSTEP is currently working with MNRE and energy departments in various states to offer integrated software solutions through which household consumers and others in 120 cities across India can check the techno-commercial viability for RTS.

PM-Surya Ghar presents a huge **opportunity for domestic** production of solar modules

Equally, the scheme presents a huge opportunity for domestic production of modules, says Gyanesh Chaudhary, chairman and managing director of Vikram Solar, a Kolkata-based solar photovoltaic module manufacturer. Recent measures to curb dumping of solar panels, especially from top exporter China, are "a very big message to Indian manufacturers like us to increase capacity", he adds. Other prominent solar panel manufacturers in India include Waaree Energies, Tata Power Solar and Adani Solar. Household rooftop systems could also potentially open up an ecosystem given the need for innovative products and technicians.

According to Neeraj Kuldeep of CEEW, the overall potential for rooftop solar in India—after factoring in consumption patterns of households and economic viability—is around 100 GW. "If we look at it from the perspective of India's 2030 goals for renewable energy, we need to push the demand," he says. This calls for a twin-pronged push on awareness and implementation, something the Muft Bijli Yojana aims to do. ■